Thoufiq Yahya

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Senior Software Architect | Lead Embedded Engineer | C++ | Qt/QML | Software-Defined Vehicle

PROFESSIONAL SUMMARY

Lead Embedded Engineer & Software Architect with over 13 years of expertise driving the architecture and delivery of embedded Linux software, voice-enabled automotive systems, and real-time HMI platforms. Proven track record leading multi-engineer teams through ambiguous problem spaces, establishing architectural standards (ADRs), and architecting low-latency IPC layers (D-Bus, Thrift, Facelift). Specialist in Modern C++, Qt6/QML, dialogue/voice agent integration, and scalable system design.

CORE COMPETENCIES

Software Architecture & System Design

- Software Architecture, Modular Architecture, Subsystem Design, Architecture Decision Records (ADRs)
- Inter-Process Communication (IPC): D-Bus, Apache Thrift, Facelift (QML IPC framework)
- Modern C++ (C++11/14/17), Object-Oriented Design, SOLID Principles, Design Patterns
- Qt Framework, Qt6, Qt Quick, QML, JavaScript, Reusable Framework Architecture
- Technical Leadership, Team Mentoring, Code Reviews, Technical Design & UML

Performance & Graphics Engineering

- OpenGL ES, GLSL, Qt Rendering Framework, GPU-Accelerated Rendering Pipelines
- Perfetto System Tracing, QML Profiler, Trace-Based Performance Analysis, GPU/Memory Optimization
- Real-Time 3D Visualization Engines, Shader Development, Engine Feasibility Evaluation (Qt vs. Unity)
- Asset Workflows (Blender, GLTF), Resource-Constrained Embedded Hardware Optimization

Platforms, Tooling & Delivery

- Embedded Linux, Linux Application Development, System Integration
- Docker Containerization, Custom Graphics/Windowing Workstation Simulation
- CMake, GCC, Cross-Compilation, GDB, Git, CI/CD Workflows, JIRA
- Test-Driven Development (TDD), GTest, GMock, Agile Methodologies
- Python, Bash, JavaScript, ActionScript, PHP

Emerging Expertise

- Board Support Package (BSP) & Board-Level Software Concepts
- Bare-Metal Programming Fundamentals
- Android Open Source Project (AOSP) Architecture & Platform Fundamentals

PROFESSIONAL EXPERIENCE

Xpheno Staffing | Embedded Software Engineer (Client: MBRDI) **December 2024 - Present**

- Own and optimize an in-house real-time 3D visualization engine for automotive infotainment and software-defined vehicle (SDV) experiences.
- Develop custom GPU-accelerated GLSL shader effects and interactive rendering features within the Qt Graphics Framework.
- Perform trace-based performance profiling using Perfetto and QML Profiler to detect bottlenecks and eliminate duplicate GPU texture allocations, significantly reducing runtime memory footprint and draw call overhead on resource-constrained embedded Linux targets.
- Conduct technical feasibility evaluation comparing Qt/QML and Unity for next-generation SDV rendering architectures; recommended Unity based on asset lifecycle management, built-in runtime optimizations, and editor workflows.
- Collaborate across UX, platform, graphics, and integration teams to translate high-level product objectives into production-ready software features.

MBRDI | Senior Software Engineer / Technical Lead **April 2016 - September 2023**

- Architected a modular Sound Dialogue System prototype for in-car voice experiences, designing a context-aware dialogue manager and integrating multi-intent NLU engines (Dialogflow, Rasa, Wit.ai, Houndify) to enable intelligent, personalized vehicle interactions.
- Led a 5-engineer team responsible for architecture, technical design, code reviews, mentoring, and end-to-end delivery of production infotainment widgets for the MBOS (Mercedes-Benz Operating System) i3 release.
- Authored Architecture Decision Records (ADRs) to establish modular UI component standards built on Qt Quick Controls 2.
- Architected and defined IPC protocols using D-Bus, Apache Thrift, and Facelift to decouple UI components from platform daemons and backend vehicle services.
- Engineered Docker-based Linux containerized execution environments with windowing and graphics stack support to simulate target automotive software on standard developer desktop workstations, accelerating CI/CD testing and developer

onboarding.
- Designed and built advanced custom UI controls, progress indicators, animated components, and shader-driven visual effects using C++, Qt6, and QML.
- Coordinated cross-functional integration efforts with platform, CI/CD, release, and QA teams across India and Germany throughout full product lifecycles.
- Led Linux platform and development environment migrations (Ubuntu 16.04 to 20.04) utilizing Bash scripting and Docker containerization.
- Enforced Test-Driven Development (TDD) best practices using GTest and GMock to increase test coverage and codebase maintainability.

L&T Technology Services | Technical Consultant (Client: Cisco NDS) **June 2014 - April 2016**

- Developed embedded set-top box television middleware and Electronic Program Guide (EPG) applications.
- Built modular UI components during the platform migration from Flash/ActionScript to Qt/QML, JavaScript, and Modern C++.
- Collaborated with Linux platform and hardware teams on IPC integration, build configuration, system responsiveness, and memory optimization on resource-constrained embedded targets.

12Play Interactive | Game Developer **November 2012 - June 2014**

- Developed real-time multiplayer educational games and interactive learning systems using ActionScript, Flash, SmartFoxServer, and PHP/MySQL.
- Architected core gameplay systems, animations, visual effects, and network communications using object-oriented design patterns.

Excel Goodies | Game Developer / Web Designer **June 2011 - January 2012**

- Developed interactive web applications and UI components in a fast-paced startup environment using ActionScript, Flash, PHP, HTML, JavaScript, and CSS.

AWARDS & HONORS

- Silver Star, Mercedes-Benz R&D, 2020
- Winner, Innovation Boot Camp, Mercedes-Benz R&D, 2019
- Star Performer, L&T Technology Services, 2015

EDUCATION

Bachelor of Information Technology **2006 - 2009**
Sadakathullah Appa College, Manonmaniam Sundaranar University, Tirunelveli, India